



COLORIDO 2K26

Complete Feature Specification & Architectural Documentation

FESTIVAL DATES

October 15-17, 2026

OFFICIAL COMPETITIONS

29 Tournaments

PLATFORM RELEASE

Version 2.0 (Production)

ARCHITECTURE

React 18 + Node +
PostgreSQL

01 Authentication & Security Features

Student Registration & Onboarding

Auth

Comprehensive multi-field student sign-up capturing verified name, collegiate email, college institution name, degree course, engineering department, graduation year, and phone number with client and server validations.

One-Click Google OAuth 2.0

Google Identity

Seamless one-tap authentication verifying student Google ID tokens using Google's official Auth Library. Auto-populates profile avatar and institutional email.

Smart "Forgot Password" System

UX Safety

Validates if an email exists in the PostgreSQL database. If non-existent, it alerts the user and smoothly redirects them to "Create Account" instead of sending ghost links. If valid, issues a time-expiring cryptographic reset token.

Stateless JWT Authorization & RBAC

Security

Employs cryptographically signed JSON Web Tokens (JWT) carried in Axios bearer request headers. Strict Role-Based Access Control separates regular students (USER) from faculty convenors (ADMIN).

Password Hashing & API Rate Limiting

Hardening

Passwords encrypted using Bcrypt with 10 salt rounds (zero plaintext credentials). Express-Rate-Limit defends login and registration routes against brute-force attacks and bot spam.

02 Public Discovery & Homepage Features

Live Festival Countdown Ticker

Real-Time

High-precision interactive countdown timer calculating days, hours, minutes, and seconds until the festival inauguration on October 15, 2026.

Dynamic Festival Stats Showcase

Metrics

Animated metric counters highlighting 29 Official Competitions, ₹5,00,000+ Prize Pool, 3 Action-Packed Days, and 5,000+ Expected Participants.

Interactive Category Matrix

Discovery

Visual preview cards with gradient glows and quick filters across Sports (9), Cultural (10), and Technical (10) domains with direct deep-links.

Campus Venue Map & Directions

Logistics

Interactive campus guide helping visiting students locate cricket grounds, indoor sports stadiums, open-air auditoriums, and high-performance coding labs.

03 Official 29 Competitions Directory (/events)

The platform manages all 29 official multi-tier competitions scheduled for October 15–17, 2026 with real-time capacity monitoring and instant keyword search:

Category	Official Events	Highlights & Team Format
Sports (9)	Cricket, Football, Basketball, Volleyball, Kabaddi, Badminton, Table Tennis, Chess, Athletics (100m)	Turf wickets, indoor stadium, official association referees, knockout to finals.
Cultural (10)	Dance Fiesta, Classical Solo, Battle of Bands, Vocal Solo, Fashion Walk, Street Play, Stand-up Comedy, Fine Arts, Short Film, Photography	Amphitheatre main stage, acoustic line arrays, jury scoring, multi-tier prelims.
Technical (10)	24H Hackathon, Code Sprint, RoboWars, Paper Presentation, Web Sprint UI/UX, AI/ML Challenge, Circuit Mania, CAD/CAM Modeling, Technical Quiz, IoT Expo	High-performance labs, industry mentor reviews, hardware kits, live jury demo.

Instant Filtering & Live Search

Fast UI

Filter by All (29), Sports (9), Cultural (10), and Technical (10) tabs or search instantly by event name, keywords, or venue without page reloading.

Dynamic Tournament Cards

Framer Motion

Animated cards with physics-based hover effects, glowing borders, prize pool badges, team size requirements (e.g. 11–15 for Cricket), and live registration capacity bars (e.g. 14 / 32 filled).

04 Tournament Deep-Dive Pages (/events/:slug)

Round-by-Round Progression Matrix

Schedules

Breaks each tournament into distinct stages: Round 1 (Qualifiers), Round 2 (Knockouts/Semis), Round 3 (Grand Finals) with exact dates (Oct 15–17), timings, match durations, and qualification criteria.

Official Organizers & Coordinator Cards

Direct Help

Dedicated contact cards for Faculty Convenor, Student Coordinator, and Official Arbiter/Jury with one-click phone call (tel:) and direct email (mailto:) buttons.

Rulebooks & Event-Specific FAQs

Guidelines

Comprehensive eligibility criteria, scoring rubrics, equipment guidelines, and accordion FAQs addressing rain contingency, sports kits, and audio submissions.

05 Registration & Dynamic QR Ticketing Pipeline

Dynamic Solo & Team Roster Builder

Booking

Adaptive registration modal supporting solo participants and multi-member squads with dynamic teammate inputs, student ID validation, and auto-populated profiles.

Zero-Duplicate Registration Guard

SQL Integrity

Enforces a database-level composite unique constraint [eventId, email] that strictly prevents students from double-booking slots or hoarding team seats.

Cryptographic SVG QR Ticket Generation

Gate Pass

Generates a globally unique alphanumeric pass code (e.g. COL26-V8R2J) and an instant dynamic SVG QR code containing encrypted validation payload for gate scanners. Includes canvas-confetti particle celebrations.

06 Multi-Day Schedule, Results & Leaderboard

3-Day Interactive Festival Timeline

/schedule

Tabbed timeline across Day 1 (Oct 15: Inauguration & Prelims), Day 2 (Oct 16: Semis & Hackathons), and Day 3 (Oct 17: Championship Finals & Valedictory) with real-time status badges (SCHEDULED, LIVE, COMPLETED).

Podium Results & Medal Leaderboard

/results

Official published scores for 1st Place (Gold), 2nd Place (Silver), and 3rd Place (Bronze). Live inter-college medal tally calculating overall points for the Festival Rolling Trophy.

07 & 08 Support Hub & Convenor / Admin Operations (/admin)

Authenticated Support & Grievance Desk

/contact

Requires student authentication before submitting questions or complaints, shielding convenors from spam bots and linking inquiries directly to the student profile.

Interactive Convenor Operations Dashboard

/admin/dashboard

Metric cards with one-click navigation: Total Events jumps to /admin/events, Total Registrations jumps to /admin/registrations, and Inquiries jumps to /admin/messages.

Animated Gate Check-In Circular Progress Gauge

Analytics

Custom animated SVG circular ring measuring real-time scanned attendee arrivals against total registrations, giving organizers an instantaneous campus gate status.

Gate Attendee Scanner & CSV/Excel Data Export

Gate Security

Live searchable roster for gate volunteers to verify attendee QR codes in real time with one click. One-tap export to CSV/Excel for attendance records and certificate printing.

